Department of Information Engineering
MSc in Computer Engineering (a.a. 2022/2023)
Computer Architecture (Prof. Cosimo Antonio Prete)

Introduction To Quantum Computing

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Quantum Computing - Theory (4 Seminars) Luciano Lenzini

Single-Qubit Quantum System

- Quantum Bit (Qubit) Concept
- The State Space Representation and Superposition of a Single Qubit States
- Bloch Sphere Picture of a Qubit
- Single-Qubit Measurement
- Single Qubit Gates

Multi-Qubit Quantum System

- The State Space Representation of *n*-Qubit System
- Entangled States
- Basic of Multi-Qubit Measurement
- Controlled Qubit Gates

Quantum Circuits

- Quantum Circuit Diagrams
- Qubit Copying Circuit? No-Cloning Theorem
- Computing the Unitary Matrix for a Given Quantum Circuit
- Clues to Universal Quantum Gates
- Quantum Computational Complexity

Quantum Parallelism and the Deutsch-Jozsa Algorithm

- The Problem: Is *f (x)* Constant or Balanced?
- Embedding *f (x)* in a Quantum Black-Box Function
- Generalized Deutsch-Jozsa Problem

Quantum Computing - Laboratory (2 Seminars) Leonardo Bacciottini

What is a quantum computer?
- No high-level API. Only gates and circuits
- Quantum programs as machine code
- Quantum noise and the NISQ era
- Can we run circuits on real quantum devices? (Yes!)

How to run quantum circuits
- Classical simulators
- Real quantum computers
- IBM Quantum Experience: a cloud of quantum devices

Introduction to Qiskit
- Frameworks for quantum software development
- Qiskit software structure
- Brief Qiskit installation guide
- How to build and draw a quantum circuit

Running circuits with Qiskit
- Backends in Qiskit
- Qiskit Aer classical simulators
- Applying noise models to simulators
- Run a circuit on an IBM-Q quantum computer

Qiskit Exercise
- Run the circuits on both simulators and IBM-Q

Class Schedule

Part I: Theory

- 14 Aprile 2023, ore 15:00-17:00, aula SI5
- 21 Aprile 2023, ore 15:00-17:00, aula SI5
- 28 Aprile 2023, ore 15:00-17:00, aula SI5
- 5 Maggio 2023, ore 15:00-17:00, aula SI5

Part II: Laboratory

- 2 maggio 2023, ore 8:30-11:30, aula A24
- 9 maggio 2023, ore 8:30-11:30, aula A24

References on Quantum Computing Textbooks

- Thomas G. Wong, *Introduction to Classical and Quantum Computing.*
- Michael Loceff, *A Course in Quantum Computing.*
- Michael A. Nielsen and Isaac L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, Cambridge, 2000.

Nobel Prize for Physics

We begin this seminar by pointing out **two outstanding facts** that should further confirm, if any were needed, that *quantum computers* and *the quantum internet* will be a tangible reality in the years to come.

- *Nobel Prize for Physics*
- *Breakthrough Prize in Fundamental Physics*

Nobel Prize for Physics

Last year's Nobel Prize in Physics was awarded to

Alan Aspect

John F. Clauser

Anton Zeilinger

with the following motivation:

for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science

Breakthrough Prize in Fundamental Physics

Furthermore, the winners of the *2023 Breakthrough Prize in Fundamental Physics* also come from the science of quantum information

Charles H. Bennett

Gilled Brassard

Davide Deutsch

Peter Shor

for their contributions in the field of quantum information

These prizes are the most obvious proof of the paramount importance that quantum computing and the quantum Internet have now taken on

Quantum Computing and Computer Architecture Course

- The information given in the seven seminars is an *integral part* of the *Computer Architecture Course*

- During the examination, you should be prepared to answer one or more questions regarding quantum computing or you will be asked to design and implement a simple quantum algorithm

Premises for the Course

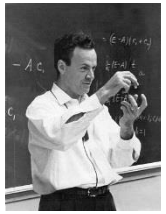
- The approaches and technologies to create the *quantum computing* are advancing so fast that the content of this cycle of lectures must inevitably be overtaken in the near future
- Here, we will introduce the most important basic concepts, while at the same time minimizing the need to understand complex physics
- With this basic grounding in the concepts, students should be able to absorb new developments in the field and assess their relative value in building complete networks

A Brief History of Quantum Computing

- In **1979**, **Paul Benioff**, a young physicist at Argonne National Labs, submitted a paper entitled *"The computer as a physical system: A microscopic quantum mechanical Hamiltonian model of computers as represented by Turing machines"*

- **Yuri Manin** also laid out the core idea of quantum computing in his **1980** book *"Computable and Non-Computable".* The book was written in Russian, however, and only translated years later

- In **1982**, **Richard Feynman** pointed out that there seemed to be essential difficulties in simulating quantum mechanical systems on classical computers, and suggested that building computers based on the principles of quantum mechanics would allow us to avoid those difficulties



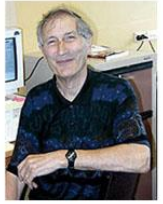
Yuri Manin Visited, as a Postdoc, the University of Pisa!!

Caro Luciano,

....

*" In breve: ho conosciuto **Yuri Manin** almeno 50 anni fa, **visitatore a Pisa** perché come **giovane dottorato** aveva già pubblicato lavori importanti per Barsotti e Bombieri(Fields Medal). Puoi averlo incontrato per caso anche tu al CNUCE o all'IEI, perché o mi veniva a trovare o perfino lavorava al CNUCE. E' diventato uno dei maggiori matematici al livello mondiale. Io avevo perso contatti con lui (troppe altre cose stupide da fare). Da pioniere ha proposto il quantum computer. E' invitato ancora in tutto il mondo; ma pare lavori a Bonn.*

Può darsi si ricordi di me; a Pisa passeggiavamo spesso assieme; forse è stato ospite a casa mia ad Asciano. Scrivigli (in inglese o tedesco se lo sai; dubito che tu conosca il russo). Se non si dà ora troppe arie passagli i miei saluti. Tanti anni fa era timidissimo. "

Gianfranco Capriz  *Pisa 26/06/2021*

Excerpt from a private email between Capriz and me.

A Brief History of Quantum Computing

- Once *Benioff*, *Manin* and *Feynman* opened the doors, other researchers (e.g., Deutsch, Jozsa, Vazirani, and Shor) began to investigate the nature of the algorithms that could run on quantum computers

A Brief History of Quantum Computing

- In 1985 - *David Deutsch* developed an algorithm which combines *quantum parallelism* with a property of quantum mechanics known as *interference*. This algorithm will be developed in the 4th seminar
- 1994 - *Peter Shor* came up with a quantum algorithm to factor very large numbers in *polynomial time*
- 1997 - *Lov Grover* develops a quantum search algorithm with $O(\sqrt{N})$ complexity
- Etc.

A Brief History of Quantum Computing

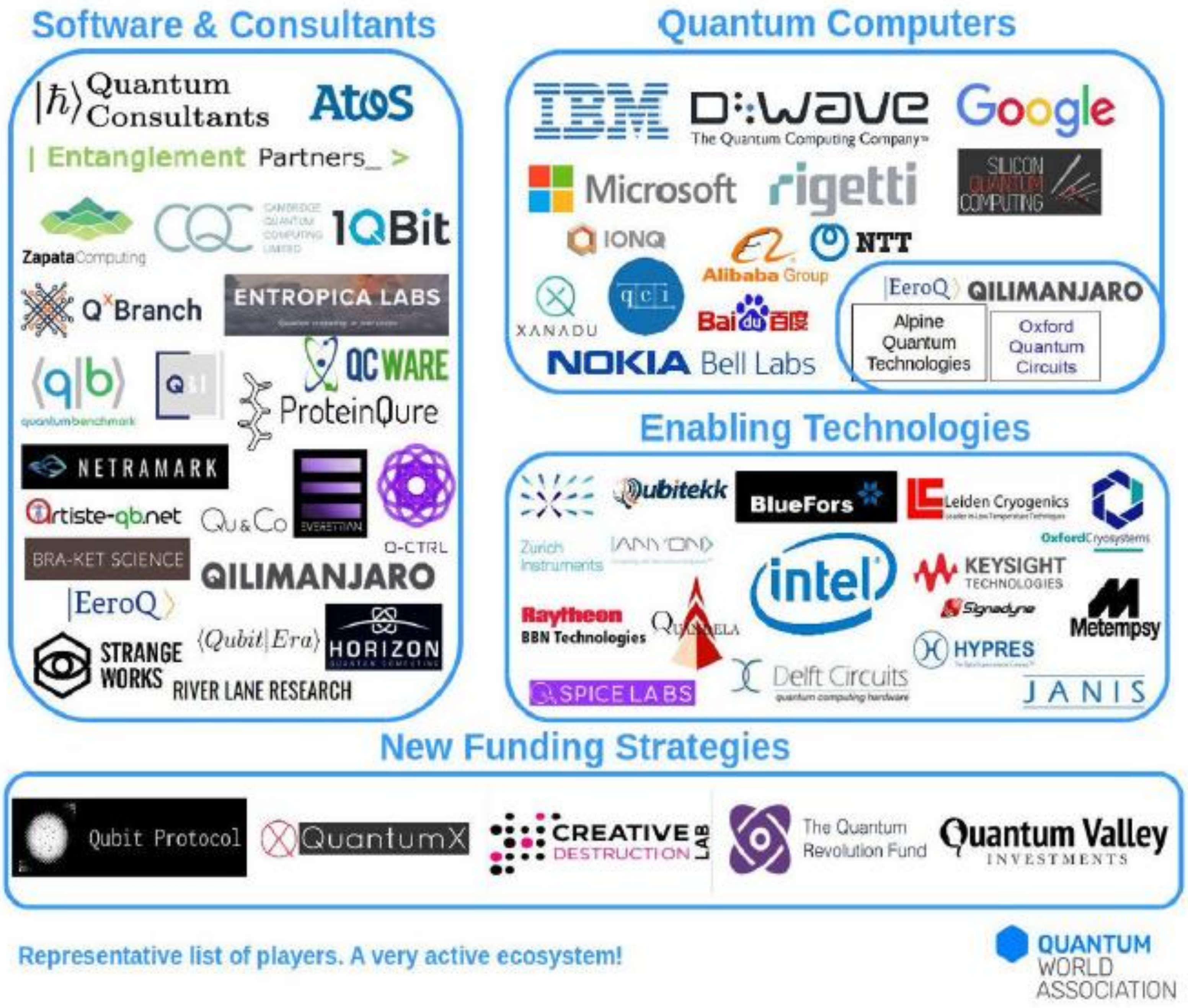
- In 1998 **Isaac Chuang** of the Los Alamos National Laboratory, **Neil Gershenfeld** of the Massachusetts Institute of Technology (MIT), and **Mark Kubinec** of the University of California at Berkeley created the first quantum computer (2-qubit) that could be loaded with data and output a solution

- Although their system was coherent for *only a few nanoseconds* and trivial from the perspective of solving meaningful problems, it demonstrated the principles of quantum computation

Large Companies Are Involved

BARCLAYS	Raytheon	IBM	intel
Google	NEC	SONY	Ford
JPMORGAN CHASE & CO.	LOCKHEED MARTIN	MITSUBISHI	HUAWEI
Microsoft	TOYOTA	AIRBUS	aws
DAIMLER	Alibaba Cloud	NOKIA	NASA
SAMSUNG	HONDA		CERN

In a Growing Ecosystem of Startups and Incumbents



IBM Quantum Development Roadmap

2023: IBM will deliver the 1,121-qubit Condor process.

2024: IBM will demonstrate a quantum system comprising three Flamingo processors totaling 1,386 qubits.

2025: IBM will demonstrate a quantum system of 3 Kookaburra processors totaling 4,158 qubits.

This leap forward will usher in a new era of scaling providing a clear path to 100,000 qubits and beyond.

Quantum Computing

- I designed the seminar cycle on the assumption that you were unfamiliar with quantum mechanics
- Therefore, I will introduce the quantum computation topics in a “soft” manner, beginning with the definition of a quantum computer

Quantum Computing

- A quantum computer is a machine that performs calculations based on the laws of *quantum mechanics* instead of the laws of *classical physics*
- A quantum computer can run applications that are beyond the reach of classical computers

What is a Qubit?

- The *quantum bit*, or *qubit* for short, is the fundamental concept of quantum computation and quantum information
- A *qubit* is similar to a classical bit in that it can take on 0 or 1 as states, but it differs from a bit in that it can also take on a continuous range of values representing a *superposition of states*

What is a Qubit?

- *A qubit is similar to a classical bit in that it can take on 0 or 1 as states*, but it differs ………

- *The first part* of the above sentence says that for a qubit two **possible** *states* are states $|0\rangle$ and $|1\rangle$, which correspond to the states 0 and 1 for a classical bit

Notation like $|\ \rangle$ is called the *Dirac notation*, which is the standard notation for states in *quantum mechanics.*

The entire object $|\bullet\rangle$ is often called a *ket*

classical bit states

qubit specific states

$$0 \rightarrow |0\rangle$$

$$1 \rightarrow |1\rangle$$

Ket Vector Representation of a Qubit

- *….. but it differs from a bit in that it can also take on a continuous range of values representing a superposition of states*

- *The second part* of the sentence states that
  - ✓ qubits are NOT constrained to be wholly $|0\rangle$ or wholly $|1\rangle$ at a given instant
  - ✓ a qubit may also exist in a *superposition*, or blend of those states **simultaneously**

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad \forall a,b \in \mathbb{C} \quad \text{such that} \quad |a|^2 + |b|^2 = 1$$

- The coefficient *a* is called the *amplitude* of the $|0\rangle$ component and the coefficient *b* is called the *amplitude* of the $|1\rangle$ component

Possible Realization of a Qubit

- A physical qubit exists at the atomic/subatomic level
- As such, its behavior is described by the laws of *quantum mechanics*
- A qubit is our *basic quantum mechanical system*
- There are many ways to implement a *physical* qubit
- Logical states $|0\rangle$ and $|1\rangle$ must be matched to the states of the physical phenomena in order to completely realize them

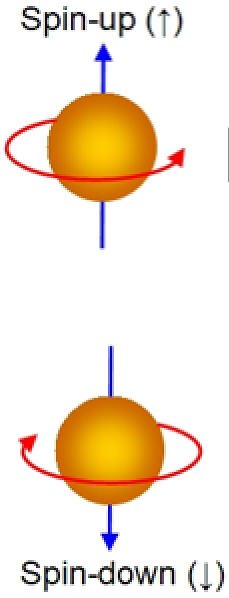
The Spin of an Electron

- When we measure the "spin" of an electron, for example, we always find it to have one of two possible values
- One value, called "spin up" or $|\uparrow\rangle$, means that the spin was found to be parallel to the axis along which the measurement was taken

classical bit states

qubit specific states

$$0 \to |0\rangle$$
$$1 \to |1\rangle$$

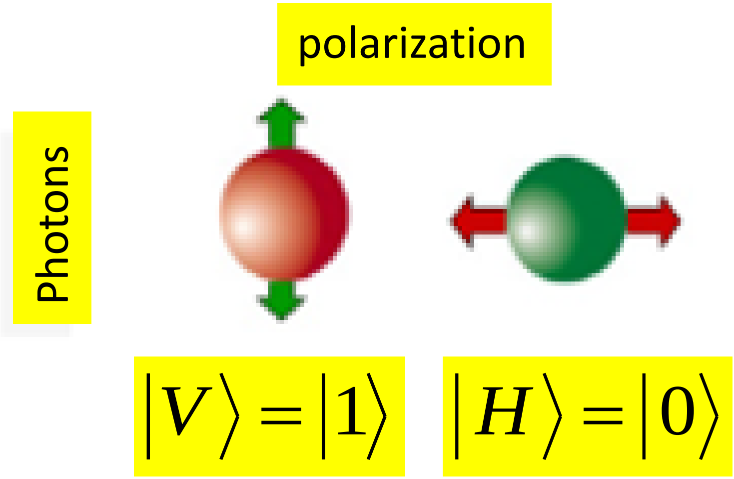


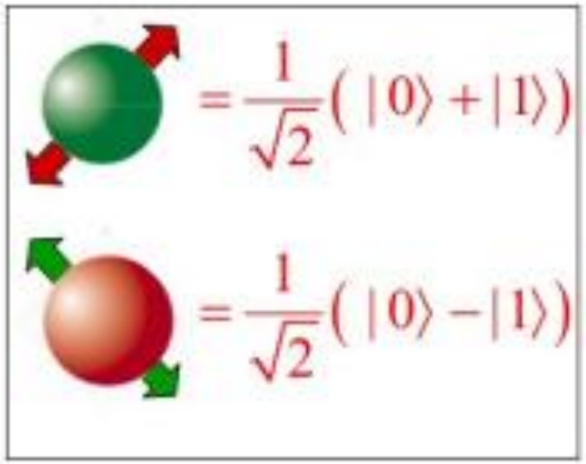
$$|0\rangle \equiv |\uparrow\rangle$$
$$|1\rangle \equiv |\downarrow\rangle$$

The other possibility, "spin-down" or $|\downarrow\rangle$, means that the spin was found to be anti-parallel to the axis along which the measurement was taken

The Photon Polarization

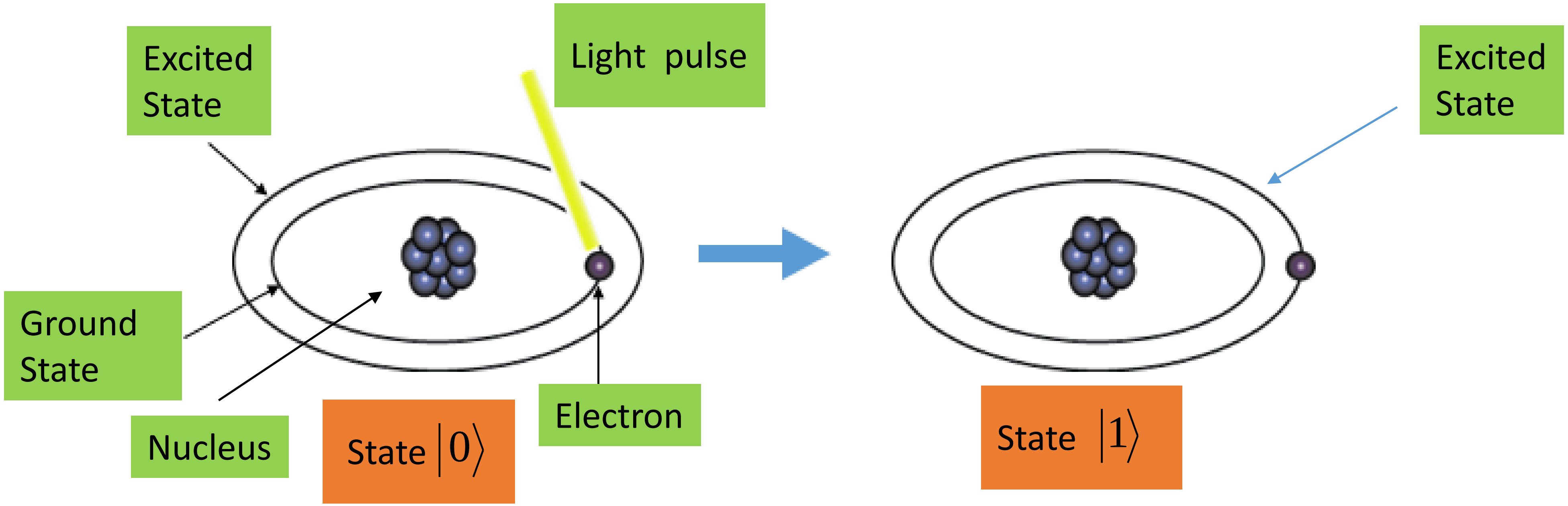
- A photon with its plane of polarization (horizontal and vertical linear polarization) could implement a physical qubit





An Excited Atom

- The discrete energy levels in an excited atom, could implement a physical qubit

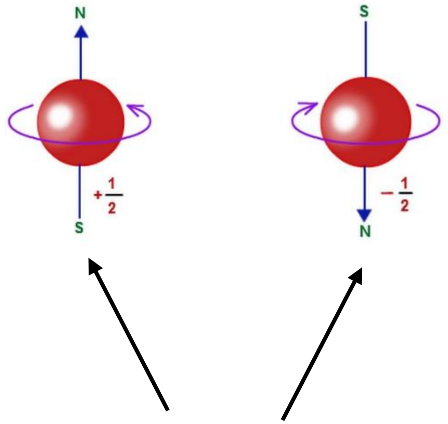


State Notation

- The notation commonly used to describe a quantum state is the *Dirac notation*
- We can put many things inside a *ket* $|\ \rangle$
- What these notations mean *physically* will depend upon the nature of the system encoding them $|\uparrow\rangle$ $|\downarrow\rangle$ $|V\rangle$ $|H\rangle$
- There are many ways to implement a *physical* qubit
- Logical states $|0\rangle$ and $|1\rangle$ must be matched to the states of the physical phenomena in order to completely realize them

Postulate 1

- **Postulate 1:** The state of a *qubit* is a *unit vector* in a *two-dimensional* **Hilbert vector space** where "special" states $|0\rangle$ and $|1\rangle$ form an *orthonormal basis* for this vector space.



$\{|0\rangle, |1\rangle\}$

Computational Basis States (CBS) or standard basis

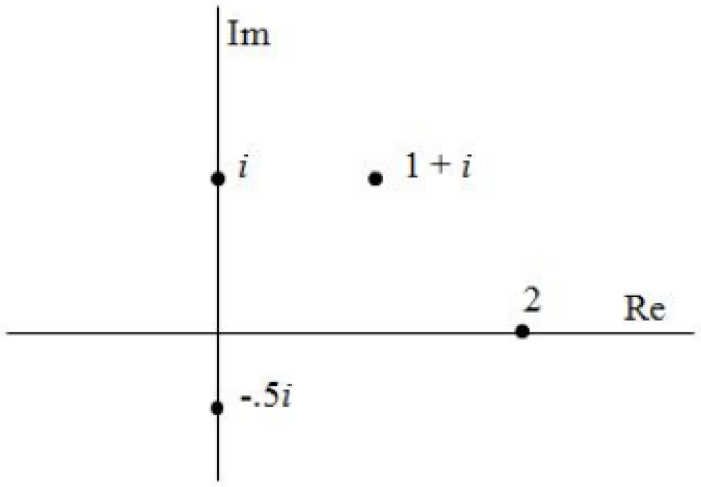
Thus, **Postulate 1** sets the arena for quantum mechanics, by specifying how the state of an isolated quantum system is to be described.

Complex Numbers

- The complex numbers are the field $\mathbb{C}$ of numbers *z* of the form *a+ib*, where *a* and *b* are real numbers and *i* is the imaginary unit equal to the square root of -1, $\sqrt{-1}$, or

$$i^2 = -1$$

- Since every complex number is defined by an ordered pair of real numbers, (*a, b*), we have a natural way to represent each such number on a plane, whose *x*-axis is the real axis (which expresses the value *a*), and *y*-axis is the imaginary axis (which expresses the value *b*)

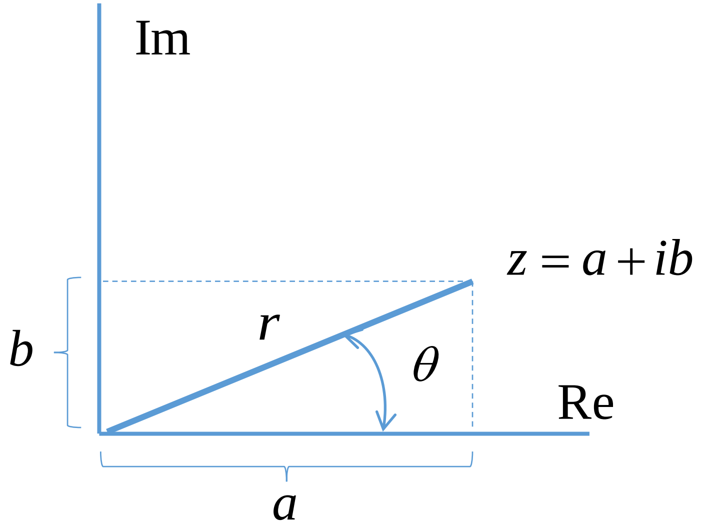


A few numbers *z=a+ib* plotted in the complex plane

Polar Representation

Besides using (a, b) to describe z, we can use the polar representation suggested by polar coordinates (r,θ) of the complex plane

$$a+ib \leftrightarrow r\left(\cos\theta+i\sin\theta\right)$$



The connection between cartesian and polar coordinates of a complex number

Terminology

- ✓ a = Re(z), the *real part* of z
- ✓ b = Im(z), the *imaginary part* of z
- ✓ r = $|z|$, the *modulus* (*magnitude*) of z
- ✓ θ = arg z , the *argument* (*polar angle*) of z
- ✓ 3i, πi, 900i, etc. *purely imaginary* numbers

Complex Conjugate

- If

$$z = a + ib$$

then its complex conjugate (or just conjugate) is designated and defined as

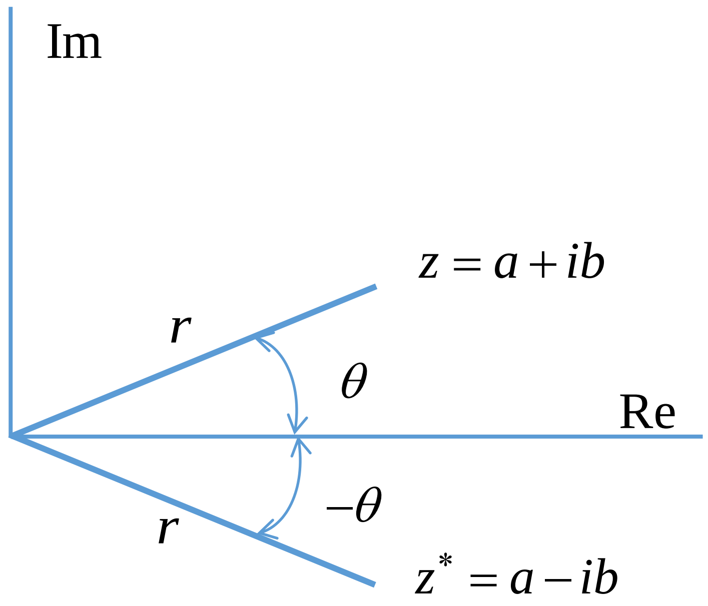
$$z^* = a - ib$$

the complex conjugate of *z*

- Furthermore

$$\left(z^*\right)^* = \left(\left(a+ib\right)^*\right)^* = \left(a-ib\right)^* = a + ib = z \longrightarrow$$

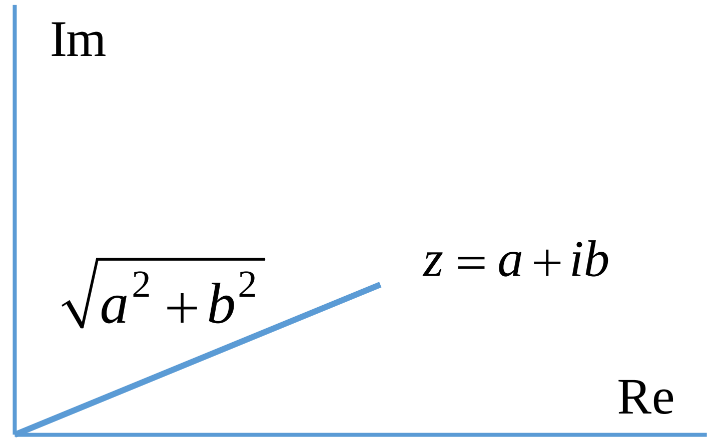
$$\left(z^*\right)^* = z$$



Conjugation as reflection

The Modulus of a Complex Number

The modulus of the complex *z* is the distance of the line segment (in the complex plane) between 0 and *z*, that is:



Modulus of a complex number

$$|z| = \sqrt{[\mathrm{Re}(z)]^2 + [\mathrm{Im}(z)]^2} = \sqrt{a^2+b^2}$$

A short computation shows that multiplying *z* by its conjugate, z^*, results in a non-negative real number which is the square of the modulus of *z*

$$|z|^2 = zz^* = z^*z = a^2 + b^2 \quad \rightarrow$$

$$|z| = \sqrt{a^2+b^2}$$

Euler's Formula

- Euler's formula

$$e^{i\theta} = \cos\theta + i\sin\theta \quad \rightarrow \quad \left|e^{i\theta}\right| = \sqrt{e^{i\theta}\cdot e^{-i\theta}} = 1,\ e^{i(\theta+\omega)} = e^{i\theta}e^{i\omega}$$

- Examples

$$e^{i\pi} = \cos\pi + i\sin\pi = -1, \qquad e^{i\frac{\pi}{2}} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i$$

- From

$$z = a + ib = r\left(\cos\vartheta + i\sin\vartheta\right) \rightarrow \quad z = re^{i\theta}$$

Hilbert Space Definition

- A **Hilbert Space** is a real or complex *vector space* that
  - has an *inner product* and
  - is *complete*

- Completeness here means that any Cauchy sequence of vectors in the space converges to some vector also in the space *(this property will not be used in my seminars)*

Review of Linear Vector Spaces

- The next three slides will be helpful as you go through the lesson at home

- A linear vector space *V* consists of certain elements $|u\rangle, |v\rangle, \ldots.$ called *vectors* together with a field of ordinary numbers (sometimes called *c-numbers*) *a*, *b*, *c*,….

- In quantum mechanics, the latter are the complex numbers, and we deal with *complex vector spaces*

- The vectors $|u\rangle, |v\rangle, \ldots.$ and the numbers *a*, *b*, *c*, … satisfy the following rules:

Review of Linear Vector Spaces

- *Vector addition is defined.* If $|u\rangle$ and $|v\rangle$ are members of *V*, there exists another vector $|w\rangle$, also a member of *V*, such that

$$|w\rangle = |u\rangle + |v\rangle$$

- *Vector addition is commutative.*

$$|u\rangle + |v\rangle = |v\rangle + |u\rangle, \qquad \forall |u\rangle, \forall |v\rangle \in V$$

- *There exists a null vector* or *zero vector* , denoted by $|zero\rangle$ or simply 0, such that

$$|u\rangle + |zero\rangle = |zero\rangle + |u\rangle = |u\rangle, \quad \forall |u\rangle \in V$$

Review of Linear Vector Spaces

- *Vector Multiplication of a vector $|u\rangle$ by any c-number a is defined*

$$|u'\rangle = a|u\rangle$$

is in the same “direction” (along the same ray) as $|u\rangle$ and

$$a|zero\rangle = |zero\rangle$$

- *The following distributive law holds*

$$a\left(|u\rangle + |v\rangle\right) = a|u\rangle + a|v\rangle \qquad \forall|u\rangle, \forall|v\rangle \in V \text{ and for any c-number } a$$

Superposition

- In the standard basis, $\{|0\rangle, |1\rangle\}$ the basis elements $|0\rangle$ and $|1\rangle$ can be expressed as

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- A superposition or complex linear combination $|\psi\rangle = a|0\rangle + b|1\rangle$ can be written

$$|\psi\rangle = a|0\rangle + b|1\rangle = a\begin{bmatrix} 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ b \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

Superposition

- The notion of **superposition** is basis-dependent; all states are superposition with respect to some bases and not with respect to others

- For instance, take this vector

$$|+\rangle = \left(|0\rangle + |1\rangle\right) / \sqrt{2}$$

- This vector is a superposition with respect to the basis $\{|0\rangle, |1\rangle\}$ but not with respect to $\{|+\rangle, |-\rangle\}$ or

$$\left\{ \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\}$$

Bras, Kets, Inner and Outer Products

- For every *ket* $|\psi\rangle$

  which can be thought of as a shorthand notation for a *column vector*

  there is a corresponding *bra* $\langle\psi|$

  which can be though of as shorthand for a *row vector*

- Mathematically, they are the dual of one another

$$|\psi\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

ket

$$\langle\psi| = a^*\langle 0| + b^*\langle 1| = \begin{bmatrix} a^*, & b^* \end{bmatrix}$$

bra

- $\langle\psi|$ is the *transposed conjugate* of $|\psi\rangle$

> *Note*
> If z=x+iy is a complex number with real part x and part y, then the complex conjugate of z is z*=x-iy

Examples

Here are some kets (not necessarily normalized) and their associated bras.

$$|\psi\rangle = \begin{pmatrix} 1 + i \\ \sqrt{2} - 2i \end{pmatrix} \longrightarrow \langle\psi| = \begin{pmatrix} 1 - i, & \sqrt{2} + 2i \end{pmatrix}$$

$$|\psi\rangle = \begin{pmatrix} \sqrt{3}/2 \\ -i \end{pmatrix} \longrightarrow \langle\psi| = \begin{pmatrix} \sqrt{3}/2, & i \end{pmatrix}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 5i \\ 0 \end{pmatrix} \longrightarrow \langle\psi| = \frac{1}{\sqrt{2}} \left(-5i, \; 0 \right)$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \longrightarrow \langle\psi| = \frac{1}{\sqrt{2}} \left(1, \; 1 \right)$$

$$|+\rangle \longrightarrow \langle+|$$

Bras, Kets, Inner and Outer Products

- The *ket* and the *bra* contain *equivalent information* about the quantum state in question

- What is the purpose of introducing *bra* vectors into the discussion if they don't contain any new information about the quantum state?

- It turns out that products of *bras* and *kets* give us insight into similarities between two quantum states

Inner Product

- For a pair of qubits in states

$$|\psi\rangle = a|0\rangle + b|1\rangle \qquad |\phi\rangle = c|0\rangle + d|1\rangle$$

we can define their *inner product*

$$\langle\psi|\phi\rangle = (\langle\psi|)\cdot(|\phi\rangle) = [a^* \quad b^*]\cdot\begin{bmatrix} c \\ d \end{bmatrix} = a^*c + b^*d$$

- Note that the result is just a number, or *scalar*
- So, an inner product is also called *scalar product*

Norm of a Vector

- We define the *norm* of a vector $|\psi\rangle$ by

$$\||\psi\rangle\| \equiv \sqrt{\langle\psi|\psi\rangle}$$

- A *unit vector* is a vector $|\psi\rangle$ such that $\||\psi\rangle\| = 1$

- We also say that $|\psi\rangle$ is *normalized* if $\||\psi\rangle\| = 1$

- It is convenient to talk of *normalizing* a vector by dividing by its norm; thus $|\psi\rangle / \||\psi\rangle\|$ is the *normalized* form of $|\psi\rangle$, for any *non-zero* vector $|\psi\rangle$

Exercise

Prove that $\langle \psi | \phi \rangle = \langle \phi | \psi \rangle^*$ where $|\psi\rangle = a|0\rangle + b|1\rangle$ and $|\phi\rangle = c|0\rangle + d|1\rangle$

Proof

$$\langle \psi | \phi \rangle = \begin{bmatrix} a^* & b^* \end{bmatrix} \cdot \begin{bmatrix} c \\ d \end{bmatrix} = a^* c + b^* d$$

$$\langle \phi | \psi \rangle = \begin{bmatrix} c^* & d^* \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} = c^* a + d^* b$$

$$\langle \phi | \psi \rangle^* = \left(c^* a + d^* b \right)^* = ca^* + db^* = a^* c + b^* d = \langle \psi | \phi \rangle$$

Inner Product

- $\langle \psi | \phi \rangle$ is a scalar which varies from *zero* for *orthogonal states* to *one* for *identical normalized states*

- For this reason, $\langle \psi | \phi \rangle$ is called the *overlap* between (normalized) states $| \psi \rangle$ and $| \phi \rangle$

Inner Product for Identical States

- When

$$|\psi\rangle = |\phi\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

their *inner product* is

$$\langle\psi|\psi\rangle = (\langle\psi|)\cdot(|\psi\rangle) = [a^* \quad b^*]\cdot\begin{bmatrix} a \\ b \end{bmatrix} = a^*a + b^*b = |a|^2 + |b|^2 = 1$$

Inner Product for Orthogonal States

- If $|\psi\rangle$ and $|\phi\rangle$ are $|0\rangle$ and $|1\rangle$

$$|\psi\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad |\phi\rangle = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Longrightarrow$$

their inner product is

$$\langle 0|1\rangle = (\langle 0|)\cdot(|1\rangle) = [1 \quad 0]\cdot\begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1\cdot 0 + 0\cdot 1 = 0 \Longrightarrow$$ $|0\rangle$ and $|1\rangle$ are orthogonal

Amplitude and Inner Product

- Given a state

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

we have

$$\langle 0|\psi\rangle = \underbrace{a\langle 0|0\rangle}_{=1} + b\underset{=0}{\langle 0|1\rangle} = a$$

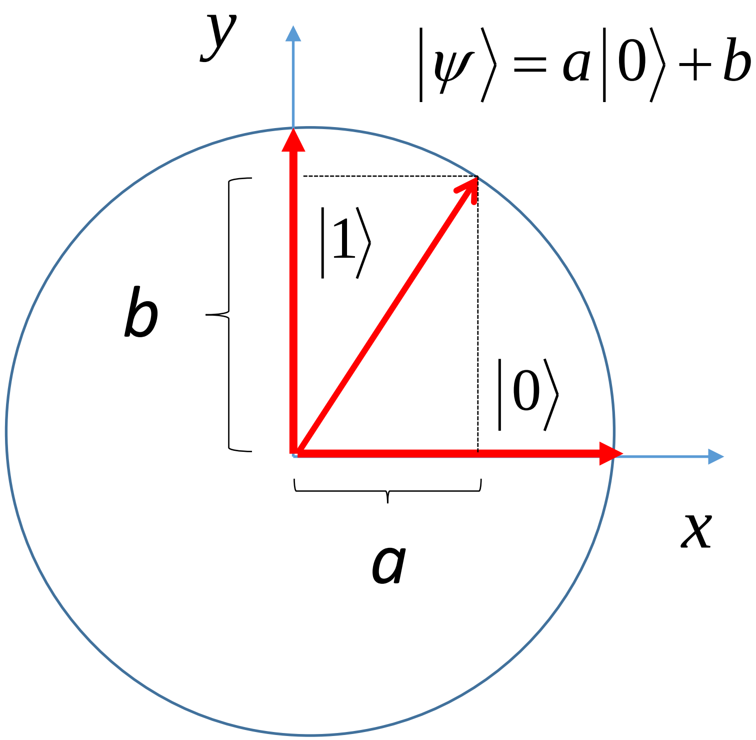
$$\langle 1|\psi\rangle = a\underset{=0}{\langle 1|0\rangle} + b\underset{=1}{\langle 1|1\rangle} = b$$

$$\longrightarrow \quad |\psi\rangle = \langle 0|\psi\rangle|0\rangle + \langle 1|\psi\rangle|1\rangle$$

Geometric Interpretation

- If $|\psi\rangle$ is a unit vector in the plane, then a and b can be interpreted as the projection of $|\psi\rangle$ along the x and y axes

$$|\psi\rangle = a|0\rangle + b|1\rangle$$



$$|0\rangle \equiv \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad |1\rangle \equiv \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

versors of the x and y axes

Orthonormality

- The two properties, *normalized* and *orthogonal*, can be combined as a single word, *orthonormal*

- So $\{|0\rangle, |1\rangle\}$ are *orthonormal* because each state is individually normalized, and they are orthogonal to each other, i.e.

$$\langle i | j \rangle = \delta_{ij}, \qquad \forall i, j \in \{0,1\}$$

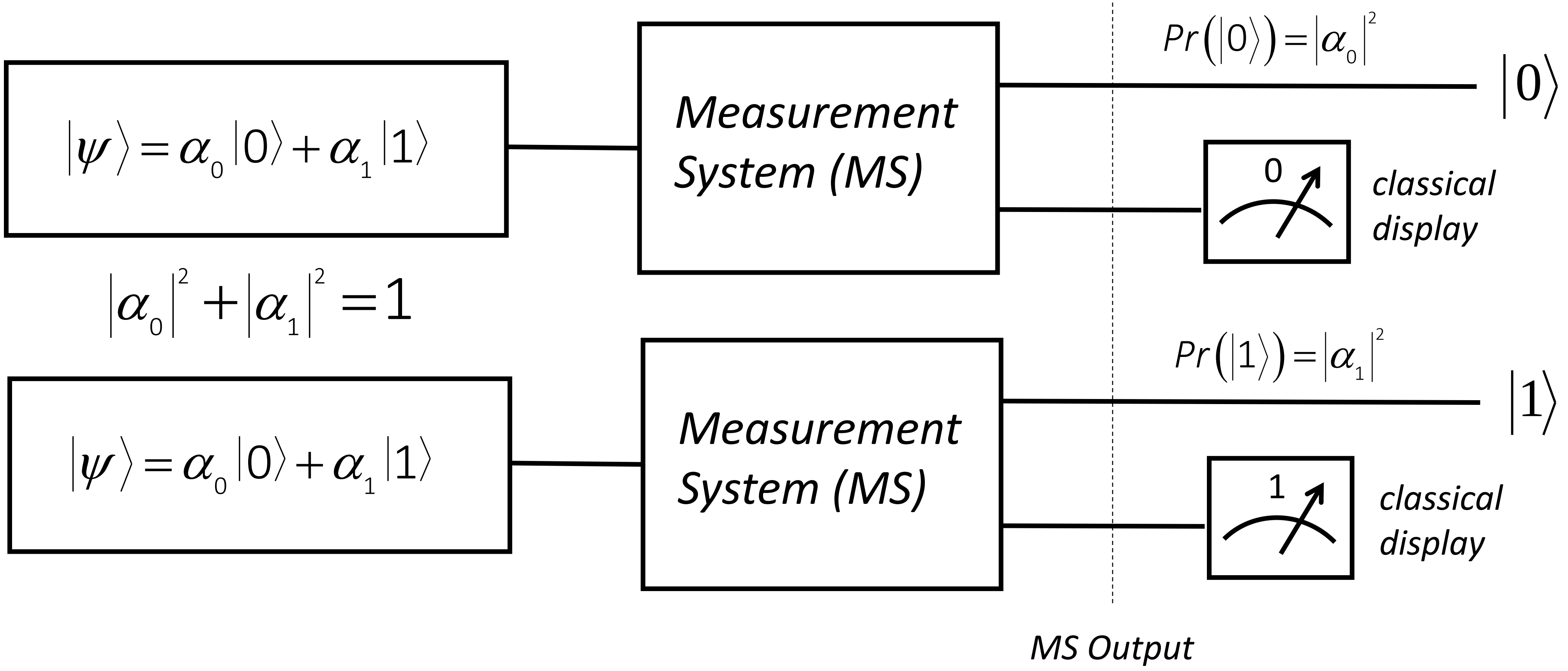
Measurement

Single Qubit Measurement

- Suppose an experimenter wants to *get information about* the state of a quantum system
- This can be obtained only by **measurement** which is carried out by a device that we call *Measurement System* or *MS*

Single Qubit Measurement

- **Postulate 3**



Single qubit measurement

- Moreover, if we *rapidly* and *repeatedly* keep measuring the same state we can suppress its evolution and effectively freeze it in a fixed quantum state

$$|\psi\rangle \xrightarrow{\text{meas.}} |0\rangle \xrightarrow{\text{meas.}} |0\rangle \xrightarrow{\text{meas.}} |0\rangle \cdots$$

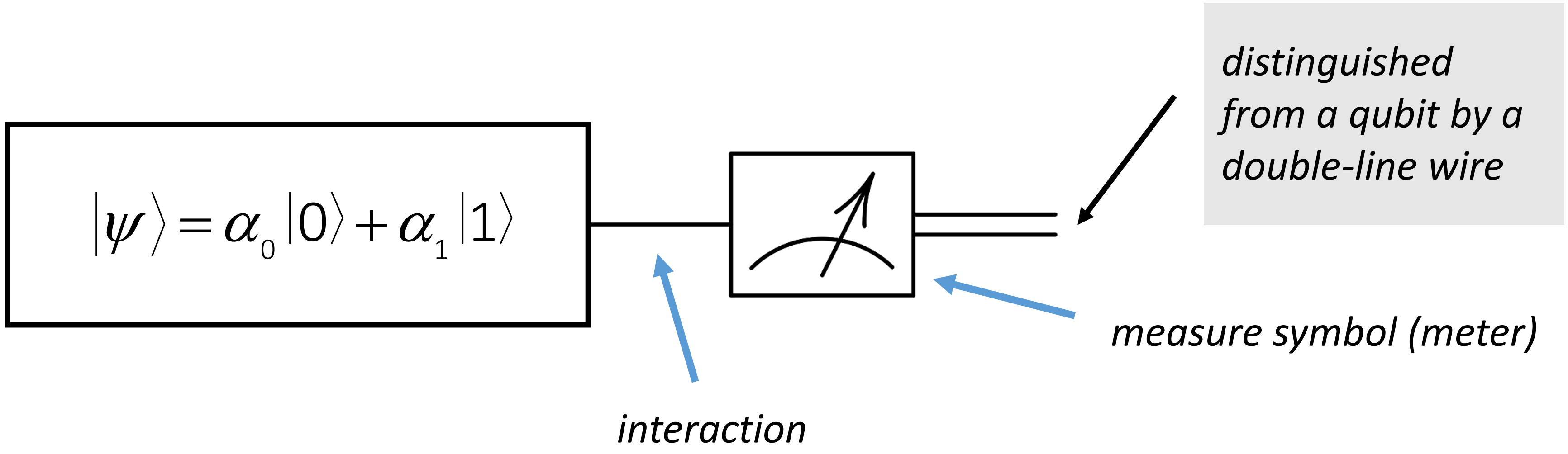
or

$$|\psi\rangle \xrightarrow{\text{meas.}} |1\rangle \xrightarrow{\text{meas.}} |1\rangle \xrightarrow{\text{meas.}} |1\rangle \cdots$$

- We said *rapidly* because if we allow time to elapse between measurements the state will, in general, evolve, or “drift off”, in accordance with Schrödinger’s equation

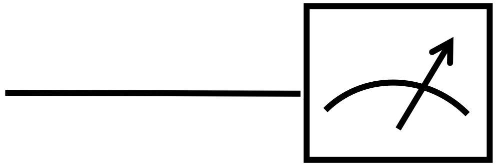
Single Qubit Measurement

- In quantum circuits, a *measurement* of a *single qubit* in the *computational basis (CBS)* is denoted by the *meter* symbol only



Single Qubit Measurement

- Quite often, the quantum outcome is discarded or ignored, and we are only interested in the classical information telling us which outcome occurred
- In such cases, we will not draw the quantum wire coming out of the measurement symbol
- We will usually omit the classical wire from circuit diagrams as well and this is the resulting symbol



Measurement Implications

1. The outcome we obtain from reading the state of a qubit in a CBS basis *is non deterministic,* i.e. sometimes we will find it in state $|0\rangle$ and sometimes we will find it in state $|1\rangle$
2. Even though a quantum bit can be in infinitely many different superposition states, it is possible to extract only *a single classical bit's worth of information* from a single quantum bit

Single Qubit Measurement

If the qubit state is

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + e^{i\pi/6}|1\rangle\right) = \frac{1}{\sqrt{2}}|0\rangle + \frac{e^{i\pi/6}}{\sqrt{2}}|1\rangle$$

to calculate these probabilities, we take the *norm-square* of the *amplitude* of $|0\rangle$ or $|1\rangle$

- That is, the probability of getting $|0\rangle$ is $\longrightarrow$ $\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

  while, the probability of getting $|1\rangle$ is $\longrightarrow$ $\left|\frac{e^{i\pi/6}}{\sqrt{2}}\right|^2 = \frac{1}{2}$

Single Qubit Measurement

- Examples

Premeasurement State (Wave function) of Qubit	Measurement Outcome	Probability of Outcome	Postmeasurement State of Qubit
$\lvert\psi\rangle = \lvert 0\rangle$	0	100%	$\lvert\psi\rangle = \lvert 0\rangle$
$\lvert\psi\rangle = \lvert 1\rangle$	1	100%	$\lvert\psi\rangle = \lvert 1\rangle$
$\lvert\psi\rangle = \frac{1}{\sqrt{2}}\lvert 0\rangle + \frac{1}{\sqrt{2}}\lvert 1\rangle$	0	50%	$\lvert\psi\rangle = \lvert 0\rangle$
	1	50%	$\lvert\psi\rangle = \lvert 1\rangle$
$\lvert\psi\rangle = \frac{1}{2}\lvert 0\rangle + \frac{\sqrt{3}}{2}\lvert 1\rangle$	0	25%	$\lvert\psi\rangle = \lvert 0\rangle$
	1	75%	$\lvert\psi\rangle = \lvert 1\rangle$
$\lvert\psi\rangle = \frac{1}{2}\lvert 0\rangle + \frac{\sqrt{3}e^{-i\pi/4}}{2}\lvert 1\rangle$	0	25%	$\lvert\psi\rangle = \lvert 0\rangle$
	1	75%	$\lvert\psi\rangle = \lvert 1\rangle$

Global Phase and Relative Phase

A general qubit state can be written

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

with complex numbers α and β, and the normalization constraint $\langle\psi|\psi\rangle = 1$ requires that:

$$|\alpha|^2 + |\beta|^2 = 1$$

We can express the state in polar coordinates as:

$$|\psi\rangle = r_\alpha e^{i\phi_\alpha}|0\rangle + r_\beta e^{i\phi_\beta}|1\rangle$$

with four real parameters r_α, ϕ_α, r_β and ϕ_β.

Global Phase and Relative Phase

Thus

$$|\psi\rangle = e^{i\phi_\alpha}\left(r_\alpha|0\rangle + r_\beta e^{i\left(\phi_\beta-\phi_\alpha\right)}|1\rangle\right)$$

or

$$|\psi\rangle = e^{i\gamma}\left(r_\alpha|0\rangle + r_\beta e^{i\phi}|1\rangle\right)$$

where $\gamma = \phi_\alpha$ *Global Phase*

$\phi = \phi_\beta - \phi_\alpha$ *Relative Phase*

Global Phase and Relative Phase

- However, the only measurement quantities are the probabilities $|\alpha|^2$ and $|\beta|^2$, so multiplying the state by an arbitrary factor $e^{i\gamma}$ (*a global phase*) has no observable consequence, because

$$\left|e^{i\gamma}\alpha\right|^2 = \left(e^{i\gamma}\alpha\right)^*\left(e^{-i\gamma}\alpha^*\right) = \alpha^*\alpha = |\alpha|^2$$

and similarly for $|\beta|^2$

Global Phase and Relative Phase

So we are free to multiply our state

$$|\psi\rangle = e^{i\gamma}\left(r_\alpha|0\rangle + r_\beta e^{i\phi}|1\rangle\right)$$

by $e^{-i\gamma}$ giving

$$|\psi'\rangle = e^{-i\gamma}|\psi\rangle = r_\alpha|0\rangle + r_\beta e^{i\phi}|1\rangle$$

which now has only three real parameters r_α, r_β, ϕ

Conclusion on Global Phase

- Therefore, from an observational point of view, $e^{i\gamma}|\psi\rangle$ and $|\psi\rangle$ are identical

- For this reason we may ignore global phase factors as being irrelevant to the observed properties of the physical system

Relative Phase

- There is another kind of phase known as the *relative phase,* which has quite a different meaning
- Consider the states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \text{and} \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Relative Phase

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \text{and} \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = \frac{|0\rangle + e^{i\pi}|1\rangle}{\sqrt{2}}$$

- For $|+\rangle$ the amplitude of $|1\rangle$ is $1/\sqrt{2}$
- For $|-\rangle$ the amplitude of $|1\rangle$ is $-1/\sqrt{2}$
- In each case the *magnitude* of the amplitudes is the same, but they differ in sign

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Relative Phase

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \text{and} \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = \frac{|0\rangle + e^{i\pi}|1\rangle}{\sqrt{2}}$$

- More generally still, *two states* are said to *differ by a relative phase* in some basis if each of the amplitudes in that basis is related by such a phase factor
- For example, the two states displayed above are the same up to a relative phase shift because the $|0\rangle$ amplitudes are identical (a relative phase factor of 1), and the $|1\rangle$ amplitudes differ only by a relative phase factor of −1

Relative Phase

- The difference between *relative phase factors* and *global phase factors* is that for relative phase the phase factors may vary from basis to basis

- *This makes the relative phase a basis-dependent concept unlike global phase*

- As a result, states which differ only by relative phases in some basis give rise to physically observable differences in measurement statistics, and it is NOT possible to regard these states as physically equivalent, as we do with states differing by a global phase factor

Relative Phase

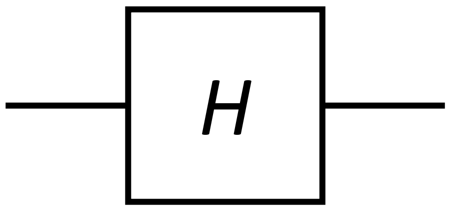
$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \text{and} \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

- In fact, if we give these states as input to the **Hadamard gate**, the state $|+\rangle$ is turned to $|0\rangle$ while the state $|-\rangle$ is turned to $|1\rangle$, which are two complete different states

Relative Phase

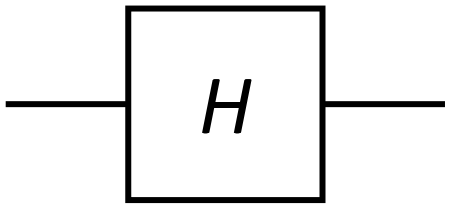
- NOTE: The Hadamard gate will be explained in the next lecture

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

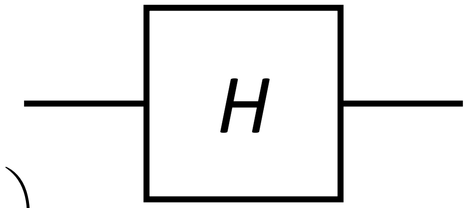


$$|0\rangle$$

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

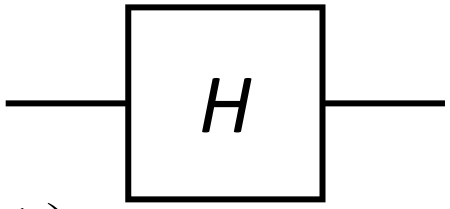


$$|1\rangle$$



$$e^{i\theta}|0\rangle$$

$$e^{i\theta}|+\rangle = e^{i\theta}\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)$$

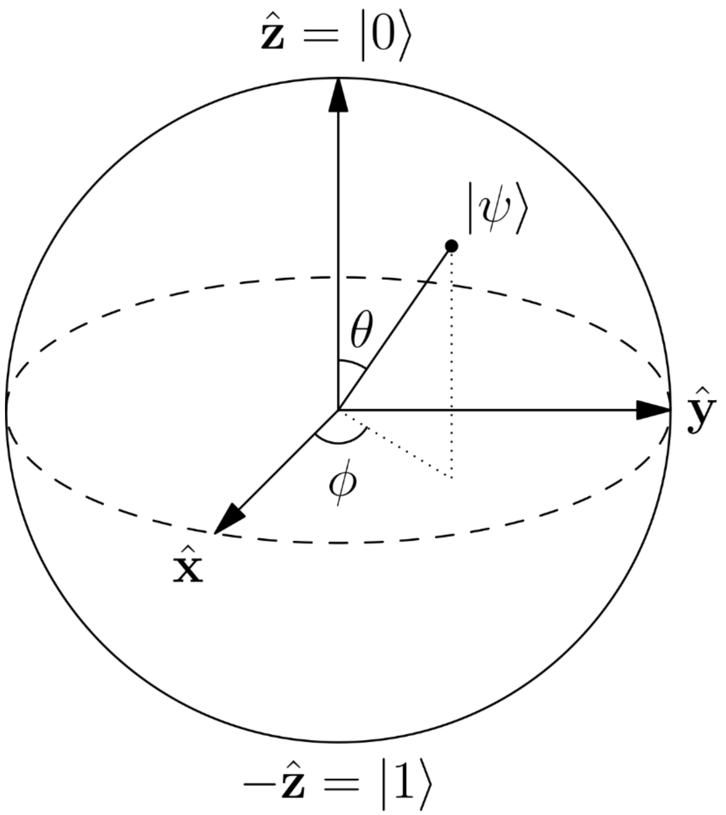


$$e^{i\theta}|1\rangle$$

$$e^{i\theta}|-\rangle = e^{i\theta}\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

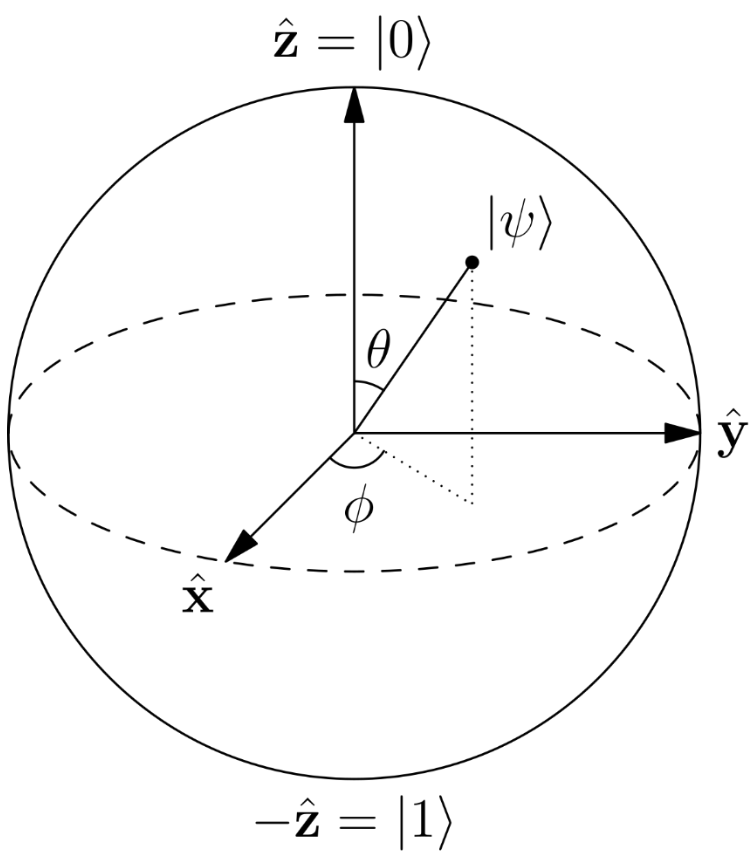
Bloch Sphere Picture of a Qubit

- A way to visualize the quantum state of a *single qubit* is to picture it as a *unit vector* inside a bounding sphere, called the *Bloch sphere*



The parameters defining the quantum state are related to the *azimuth* (ϕ) and *elevation* (θ) angles that determine where the tip of this vector touches the surface of the Bloch sphere

Bloch Sphere Picture of a Qubit



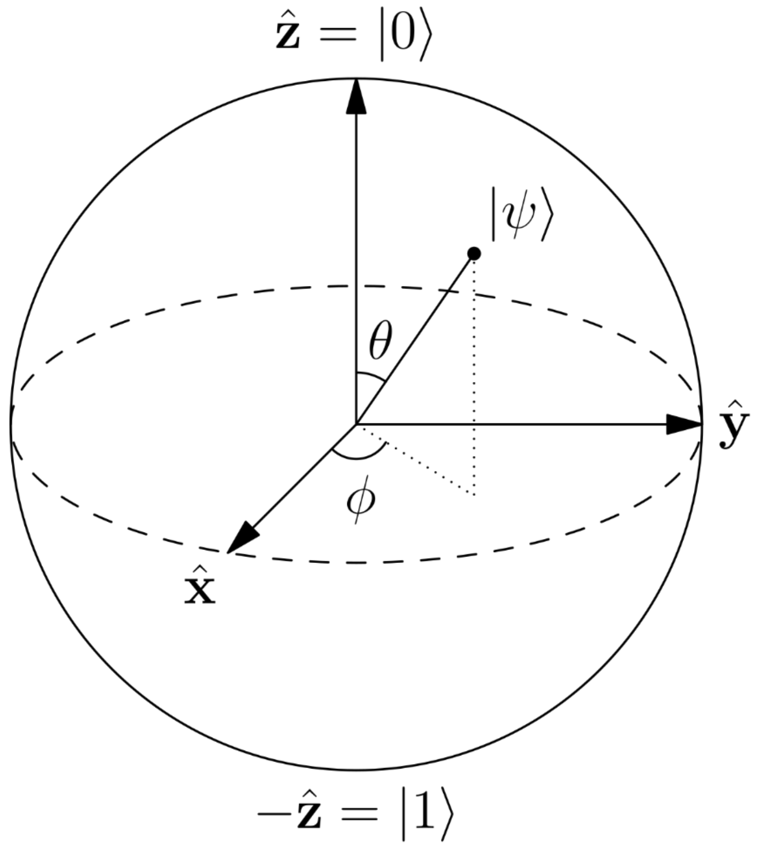
- An arbitrary state of a single qubit $|\psi\rangle = a|0\rangle + b|1\rangle$ such that $|a|^2 + |b|^2 = 1$ can be written in terms of these *azimuth* and *elevation* angles as:

$$|\psi\rangle = e^{i\gamma}\left(\cos\frac{\vartheta}{2}|0\rangle + e^{i\phi}\sin\frac{\vartheta}{2}|1\rangle\right)$$

where γ, θ, and ϕ are all real number

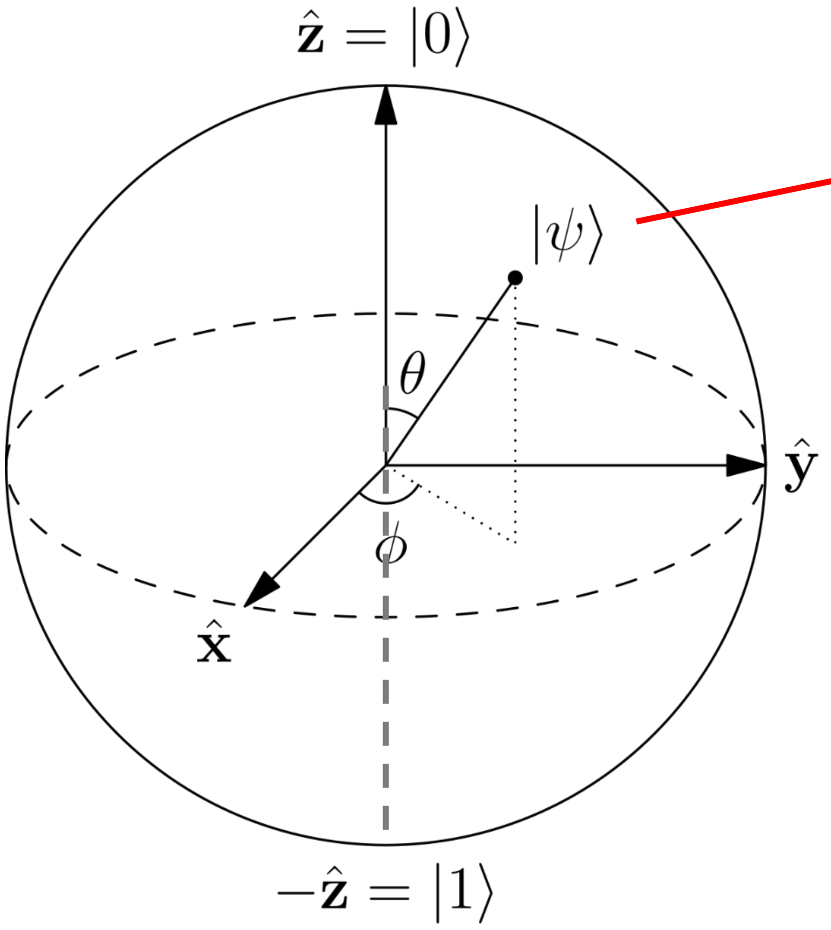
- A pair of elevation and azimuth angles (θ, ϕ) in the range $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$ pick out a point on the Bloch sphere

Bloch Sphere Picture of a Qubit



- Qubit states corresponding to different values of γ are indistinguishable and are all represented by the same point on the Bloch sphere

Bloch Sphere Picture of a Qubit



$$|\psi\rangle = e^{i\gamma}\left(\cos\frac{\vartheta}{2}|0\rangle + e^{i\phi}\sin\frac{\vartheta}{2}|1\rangle\right)$$

- The *North pole* ($\theta = 0$) corresponds to the state $|0\rangle$
- The *South pole* ($\theta = \pi$) corresponds to the *(orthogonal)* state $|1\rangle$

Bloch Sphere Picture of a Qubit

Two questions:

- First, how come the *azimuth* and *elevation* angles are expressed in *half-angles?* Why not use the following expression which seems the more appropriate one.

$$|\psi\rangle = e^{i\gamma}\left(\cos\vartheta|0\rangle + e^{i\phi}\sin\vartheta|1\rangle\right)$$

- Second, how come orthogonal states are not at right angles on the Bloch sphere? Instead they are 180° apart

Bloch Sphere Picture of a Qubit

- Consider the general quantum state

$$|\varphi\rangle = a|0\rangle + b|1\rangle, \quad a,b \in \mathbb{C}, \quad |a|^2 + |b|^2 = 1$$

- Since *a* and *b* are complex numbers they can be written in either Cartesian or Polar coordinates:

$$a = x_a + iy_a = r_a e^{i\phi_a}, \quad b = x_b + iy_b = r_b e^{i\phi_b}$$

with $i = \sqrt{-1}$ and the *x*'s, *y*'s, *r*'s, and ϕ's are all real numbers

Bloch Sphere Picture of a Qubit

- Write the general state of a qubit $|\varphi\rangle = a|0\rangle + b|1\rangle$, as

$$|\varphi\rangle = r_a e^{i\phi_a}|0\rangle + r_b e^{i\phi_b}|1\rangle = e^{i\phi_a}\left(r_a|0\rangle + r_b e^{i(\phi_b - \phi_a)}|1\rangle\right) = e^{i\phi_a}\left(r_a|0\rangle + r_b e^{i\phi}|1\rangle\right)$$

where $\phi = \phi_b - \phi_a$

- Since an overall phase factor has no observable consequence, the above state is equivalent to

$$|\varphi\rangle = r_a|0\rangle + r_b e^{i\phi}|1\rangle$$

Bloch Sphere Picture of a Qubit

- Switching back to Cartesian coordinates for the amplitude of the $|1\rangle$ component we can write this state as

$$|\varphi\rangle = r_a|0\rangle + (x+iy)|1\rangle$$

- Applying normalization we have $|r_a|^2 + |x+iy|^2 = 1$ or equivalently

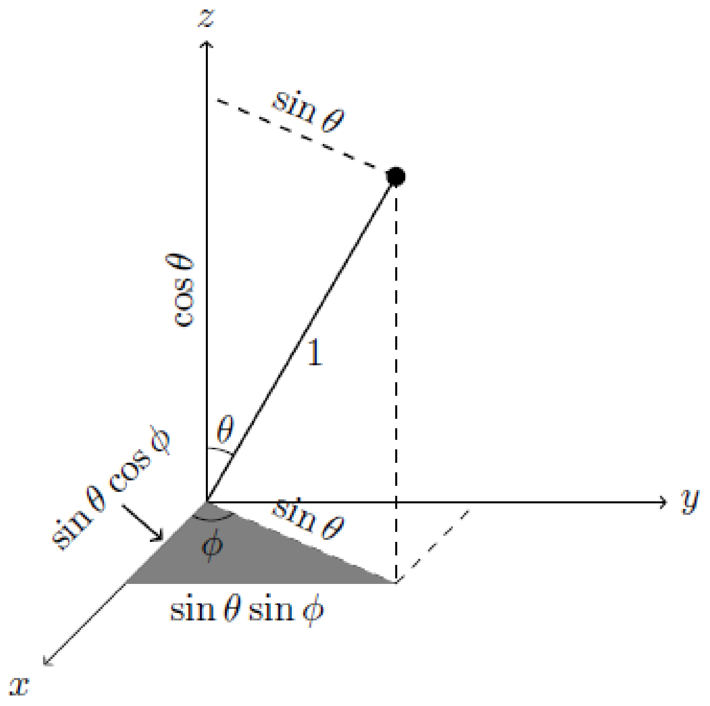
$$|r_a|^2 + (x+iy)(x+iy)^* = |r_a|^2 + (x+iy)(x-iy) = r_a{}^2 + x^2 + y^2 = 1$$

which is the equation of a sphere in coordinates r_a, *x*, and *y*

- We can rename $r_a = z$ for aesthetic reasons and it doesn't change anything but now we have the equation of a sphere in coordinates *x*, *y*, and *z,* i.e. $z^2 + x^2 + y^2 = 1$

Bloch Sphere Picture of a Qubit

- Ok so let's switch from these Cartesian coordinates to spherical coordinates. We have,



$$x = r\sin(\theta)\cos(\phi)$$
$$y = r\sin(\theta)\sin(\phi)$$
$$z = r\cos(\theta)$$

- Since *r*=1, the position on the surface of the sphere is specified using only two parameters, ϑ and φ and the general qubit state can be written

Bloch Sphere Picture of a Qubit

$$\left|\varphi\right\rangle = z\left|0\right\rangle + \left(x+iy\right)\left|1\right\rangle = \cos(\theta)\left|0\right\rangle + \left(\sin(\theta)\cos(\phi) + i\sin(\theta)\sin(\phi)\right)\left|1\right\rangle$$

$$= \cos(\theta)\left|0\right\rangle + \sin(\theta)\underbrace{\left(\cos(\phi) + i\sin(\phi)\right)}_{e^{i\phi}}\left|1\right\rangle = \cos(\theta)\left|0\right\rangle + \sin(\theta)e^{i\phi}\left|1\right\rangle$$

$$\Longrightarrow$$

$$\left|\varphi\right\rangle = \cos(\theta)\left|0\right\rangle + \sin(\theta)e^{i\phi}\left|1\right\rangle$$

Why Half Angles

But this is still not the Bloch sphere. What about the half angles?
Let

$$|\psi\rangle = \cos\theta'|0\rangle + e^{i\phi}\sin\theta'|1\rangle$$

and notice that $\theta' = 0 \Rightarrow |\psi\rangle = |0\rangle$ and $\theta' = \frac{\pi}{2} \Rightarrow |\psi\rangle = e^{i\phi}|1\rangle$

which suggests that $0 \leq \theta' \leq \frac{\pi}{2}$ may generate all points on the Bloch sphere.

Why Half Angles

Consider a state $|\psi'\rangle$ corresponding to the *opposite* point on the sphere, which has polar coordinates $(1, \pi - \theta', \phi + \pi)$

$$\begin{aligned}
|\psi'\rangle &= \cos(\pi - \theta')|0\rangle + e^{i(\phi+\pi)} \sin(\pi - \theta')|1\rangle \\
&= -\cos(\theta')|0\rangle + e^{i\phi}e^{i\pi} \sin(\theta')|1\rangle \\
&= -\cos(\theta')|0\rangle - e^{i\phi} \sin(\theta')|1\rangle \\
|\psi'\rangle &= -|\psi\rangle = e^{i\pi}|\psi\rangle,
\end{aligned}$$

$$e^{i\pi} = \cos\pi + i\sin\pi = -1$$

So it is only necessary to consider the upper hemisphere $0 \leq \theta' \leq \frac{\pi}{2}$, as opposite points in the lower hemisphere differ only by a phase factor of -1 and so are equivalent in the Bloch sphere representation.

Why Half Angles

We can map points on the upper hemisphere onto points on a sphere by defining

$$\theta = 2\theta' \quad \Rightarrow \quad \theta' = \frac{\theta}{2}$$

and we now have

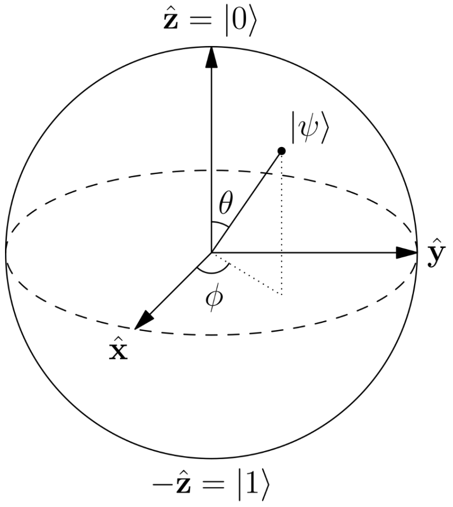
$$\boxed{|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle}$$

where $0 \le \theta \le \pi$, $0 \le \phi < 2\pi$ are the coordinates of points on the *Bloch sphere.*

Why Half Angles

We can map points on the upper hemisphere onto points on a sphere by defining

$$\theta = 2\theta' \quad \Rightarrow \quad \theta' = \frac{\theta}{2}$$



and we now have

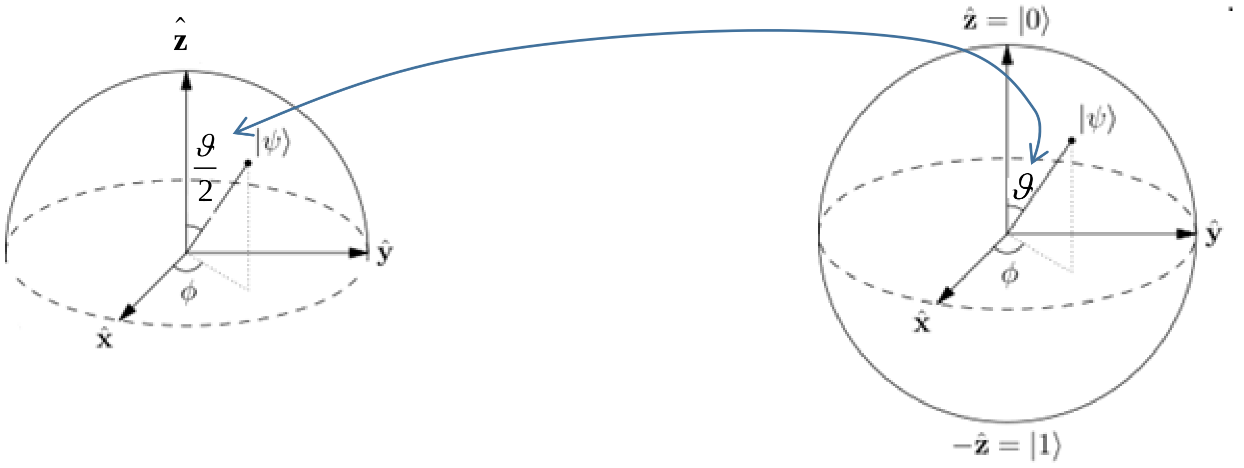
$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

where $0 \leq \theta \leq \pi$, $0 \leq \phi < 2\pi$ are the coordinates of points on the *Bloch sphere.*

Why Half Angles

$\mathbb{R}^3$

Bloch Sphere



Why Half Angles

- Notice that $\theta = 2\theta'$ is a one-to-one mapping except at $\theta' = \frac{\pi}{2}$, where all the points on the θ' 'equator' are mapped to the single point $\theta = \pi$, the 'south pole' on the Bloch sphere
- This is okay, since at the south pole $|\psi\rangle = e^{i\phi}|1\rangle$ and ϕ is a global phase with no significance. (Longitude is meaningless at a pole!)

Second Question/Orthogonality of Opposite Points

Consider a general qubit state $|\psi\rangle$

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

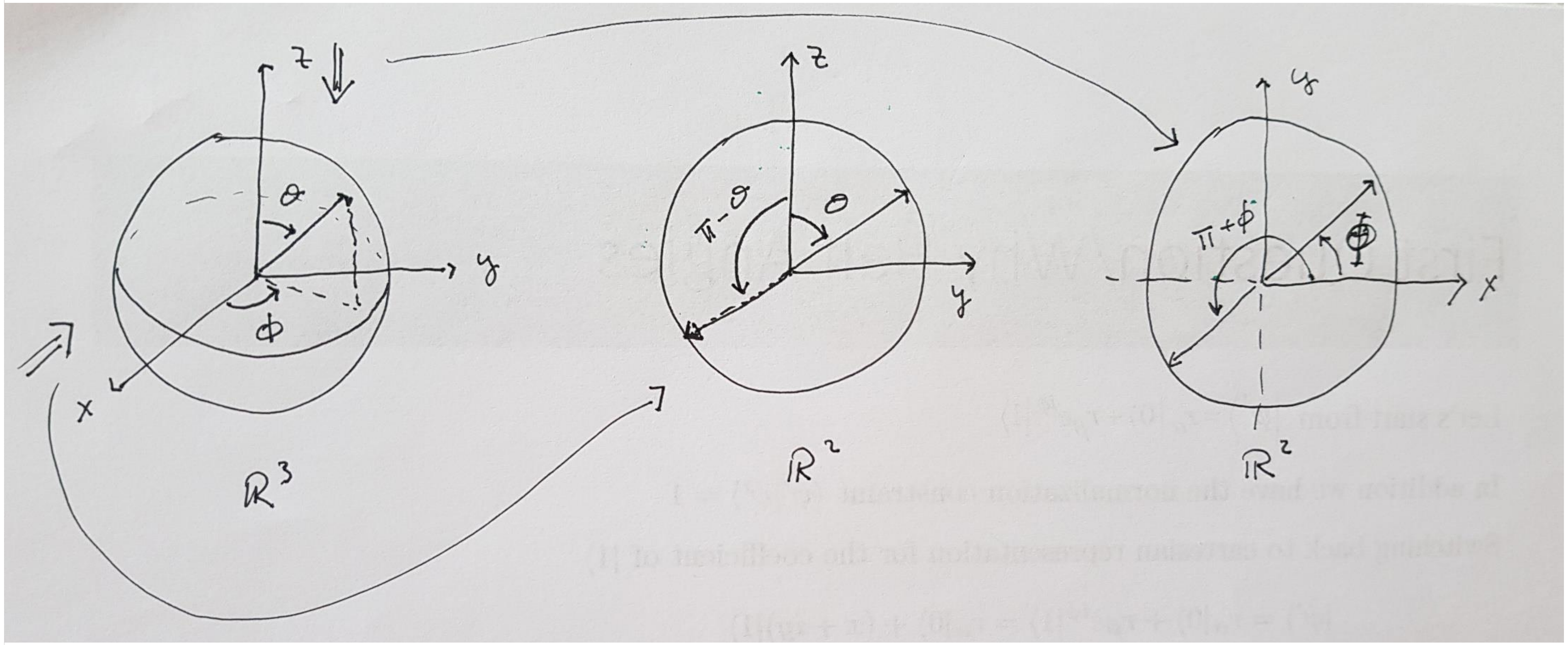
and $|\chi\rangle$ corresponding to the opposite point on the Bloch sphere

$$\begin{aligned} |\chi\rangle &= \cos\left(\frac{\pi-\theta}{2}\right)|0\rangle + e^{i(\phi+\pi)}\sin\left(\frac{\pi-\theta}{2}\right)|1\rangle \\ &= \cos\left(\frac{\pi-\theta}{2}\right)|0\rangle - e^{i\phi}\sin\left(\frac{\pi-\theta}{2}\right)|1\rangle \end{aligned}$$

So

$$\langle\chi|\psi\rangle = \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\pi-\theta}{2}\right) - \sin\left(\frac{\theta}{2}\right)\sin\left(\frac{\pi-\theta}{2}\right)$$

Second Question/Orthogonality of Opposite Points



Orthogonality of Opposite Points

$$\langle\chi|\psi\rangle = \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\pi-\theta}{2}\right) - \sin\left(\frac{\theta}{2}\right)\sin\left(\frac{\pi-\theta}{2}\right)$$

But $\cos(a+b) = \cos a \cos b - \sin a \sin b$, so

$$\langle\chi|\psi\rangle = \cos\frac{\pi}{2} = 0$$

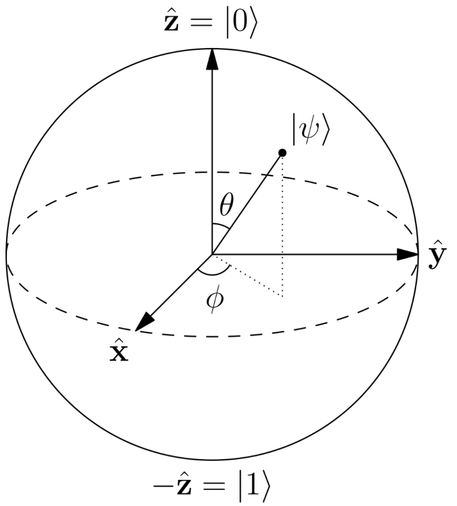
and opposite points correspond to orthogonal qubit states.

Note that in the coordinate system we used in the derivation of the Bloch sphere, with $\theta' = \theta/2$, the two points are also orthogonal - 90° apart.

Example #1

For example, let's represent on the Bloch sphere the state

$$\underbrace{|+\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)}$$

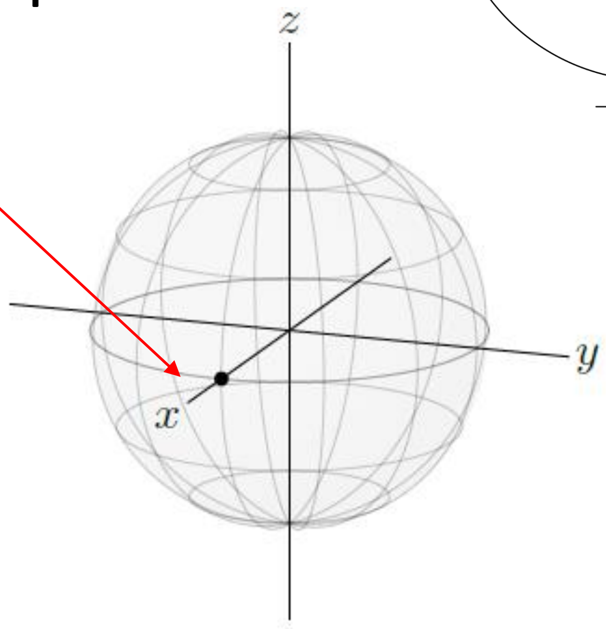


We saw erlier that a generic state on the Bloch sphere can be expressed as

$$|+\rangle = \cos\frac{\vartheta}{2}|0\rangle + e^{i\phi}\sin\frac{\vartheta}{2}|1\rangle$$

From which it follows

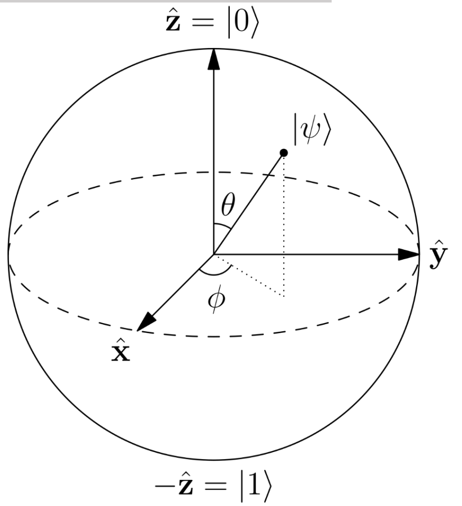
$$\boxed{\begin{gathered}\cos\frac{\vartheta}{2} = \sin\frac{\vartheta}{2} = \frac{1}{\sqrt{2}} \\ e^{i\phi} = 1\end{gathered}} \longrightarrow \boxed{\begin{gathered}\frac{\vartheta}{2} = \frac{\pi}{4} \rightarrow \vartheta = \frac{\pi}{2} \\ \phi = 0\end{gathered}} \longrightarrow$$



Example #2

For example, let's represent on the Bloch sphere the state

$$\underbrace{|-\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle - |1\rangle\right)}$$

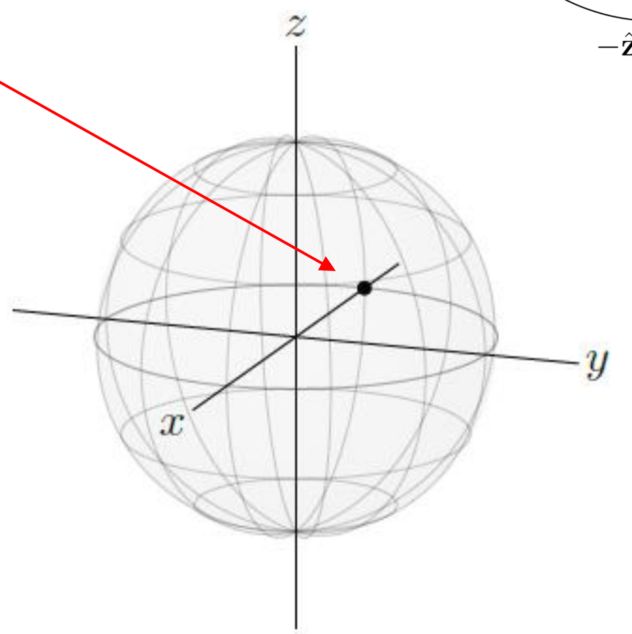


Let's compare the above state with

$$|-\rangle = \cos\frac{\vartheta}{2}|0\rangle + e^{i\phi}\sin\frac{\vartheta}{2}|1\rangle$$

from which it follows

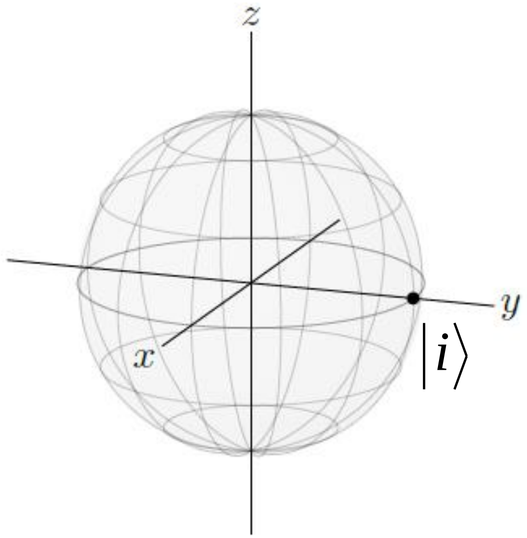
$$\boxed{\cos\frac{\vartheta}{2} = \sin\frac{\vartheta}{2} = \frac{1}{\sqrt{2}} \qquad e^{i\phi} = -1} \longrightarrow \boxed{\frac{\vartheta}{2} = \frac{\pi}{4} \rightarrow \vartheta = \frac{\pi}{2} \qquad \phi = \pi} \longrightarrow$$



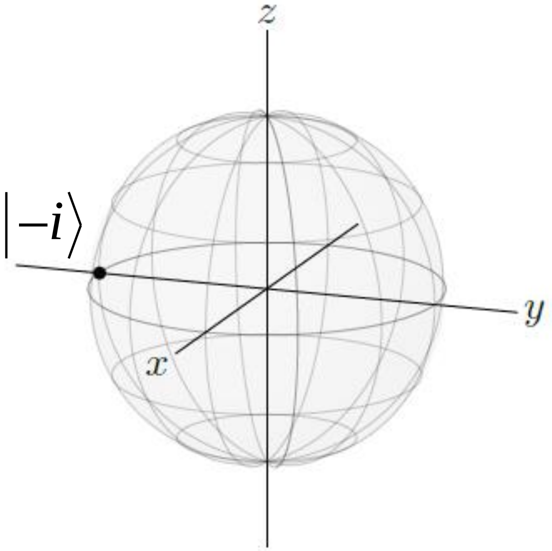
Examples #3 & #4

Two more examples

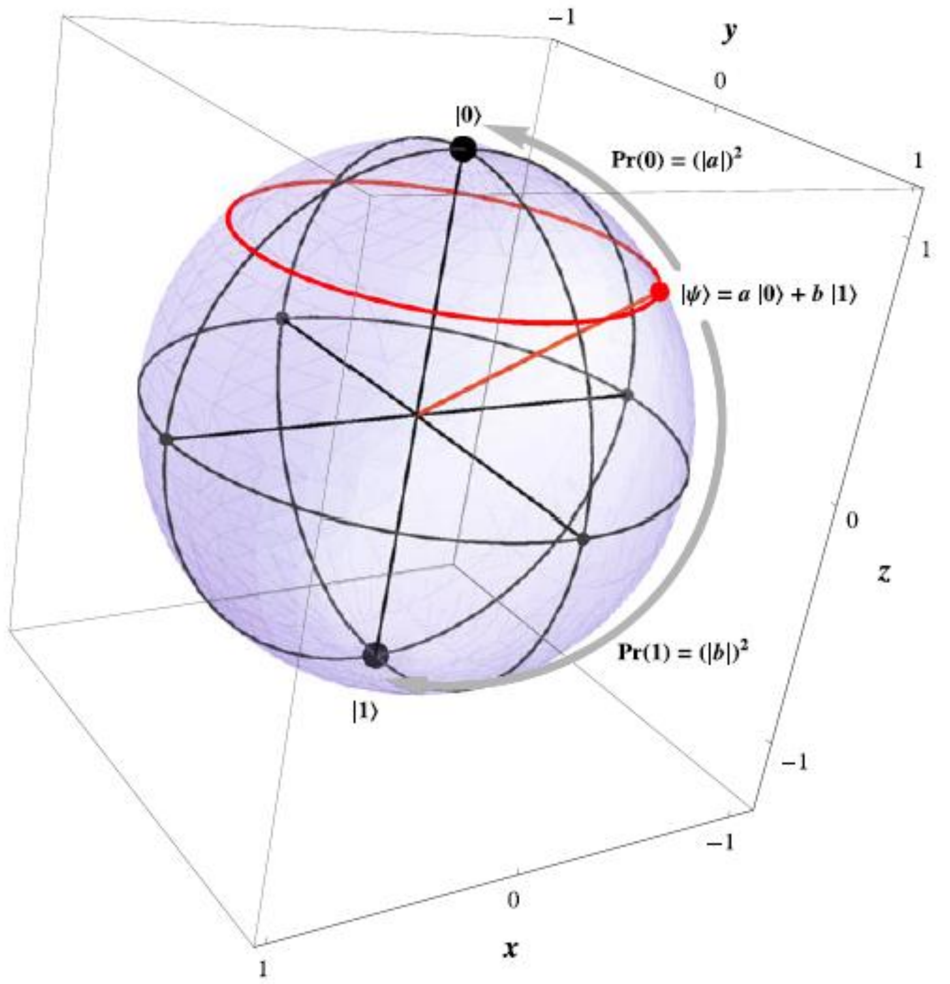
$$|i\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + i|1\rangle\right)$$



$$|-i\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle - i|1\rangle\right)$$



Measurement and the Bloch Sphere



Measurements in Other Basis

- The states $|0\rangle$ and $|1\rangle$ correspond to the north and south poles of the Bloch sphere respectively, and the axis passing through these points is the *z*-axis

- However, any two *opposite points* such as

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}};\ |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad \text{or} \quad |i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}};|-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

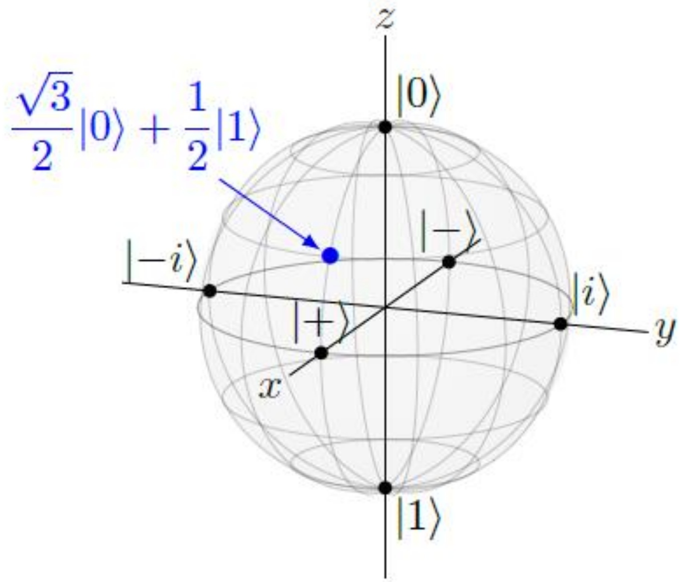
could be the *north* and *south poles*, or *any opposite points*

Measurements in Other Basis

- A set of distinct measurement outcomes is called a *basis*, and $\{|0\rangle, |1\rangle\}$ is called the *Z-basis* because they lie on the *z-axis* of the Bloch sphere

- Similarly, $\{|+\rangle, |-\rangle\}$ is called the *X-basis* because they lie on the *x-axis* of the Bloch sphere, and $\{|i\rangle, |-i\rangle\}$ is called the *Y-basis* because they lie on the *y-axis* of the Bloch sphere

- We can measure with respect to any of these bases, or with respect to any two states on opposite sides of the Bloch sphere

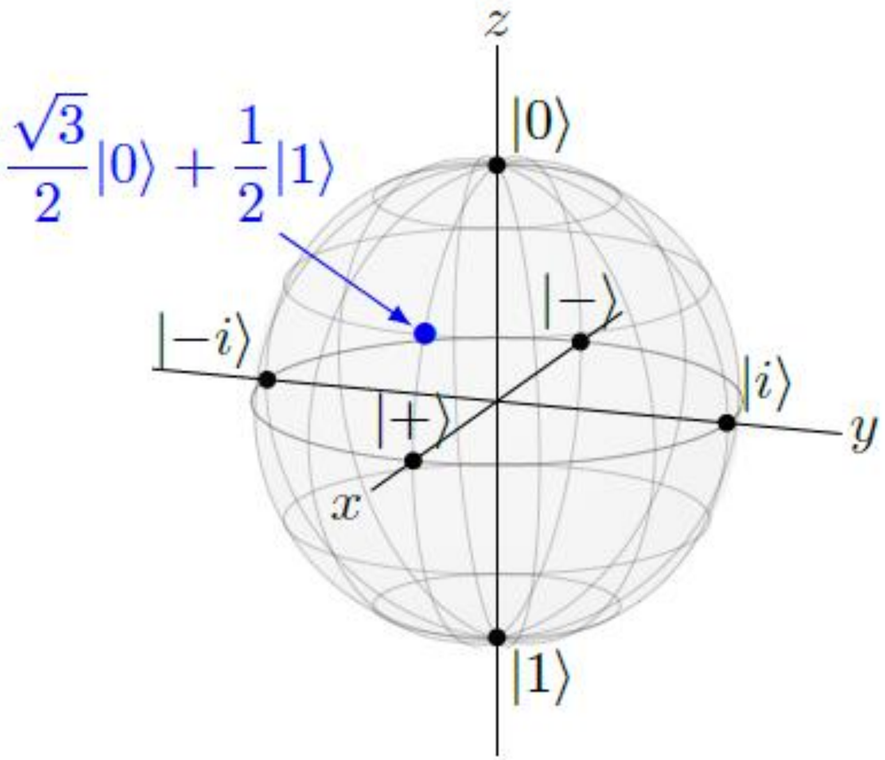


Measurements in Other Basis

- Let's consider a qubit on the state $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$

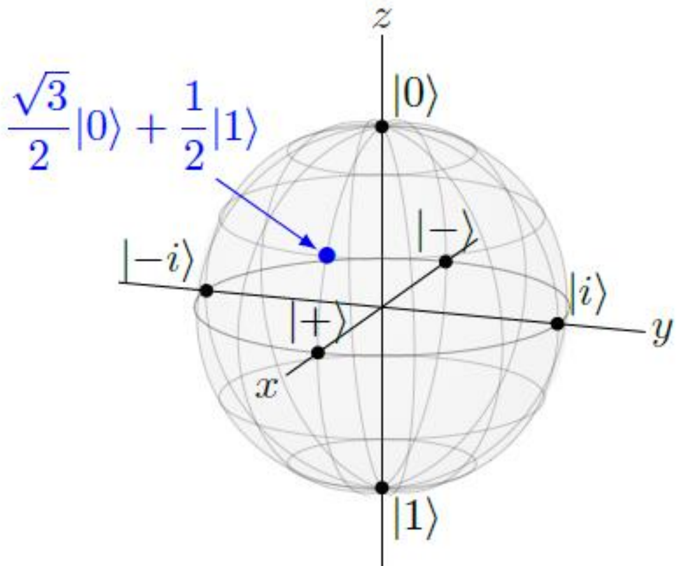
- Let us measure this with respect to four different bases
  - $\rightarrow$ the Z-basis $\{|0\rangle, |1\rangle\}$
  - $\rightarrow$ the X-basis $\{|+\rangle, |-\rangle\}$
  - $\rightarrow$ the Y-basis $\{|i\rangle, |-i\rangle\}$, and
  - $\rightarrow$ a fourth basis

$$\left\{|a\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle, |b\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle\right\}$$



Measurements in Other Basis

- If we measure the qubit in the Z-basis $\{|0\rangle, |1\rangle\}$, then we get $|0\rangle$ with probability 3/4 or $|1\rangle$ with probability 1/4
- What if we measure in the X-basis $\{|+\rangle, |-\rangle\}$ instead?
- To calculate the probabilities precisely, we need to express the state $|\psi\rangle$ in terms of $|+\rangle$ and $|-\rangle$ so that we can identify the amplitudes and then find their norm-squares



Measurements in Other Basis

- From the definitions of $|+\rangle$ and $|-\rangle$, we have

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}};\ |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle), \quad |1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle).$$

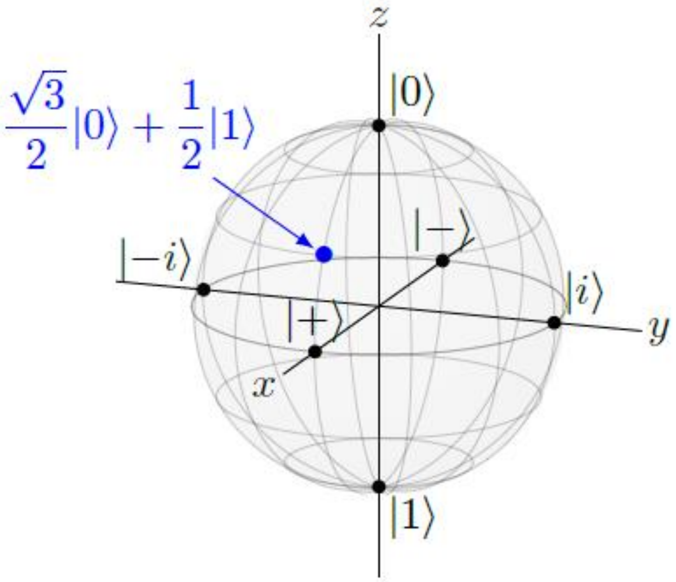
Substituting into the state of our qubit,

$$\begin{aligned}\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle &= \frac{\sqrt{3}}{2}\frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) + \frac{1}{2}\frac{1}{\sqrt{2}}(|+\rangle - |-\rangle) \\ &= \frac{\sqrt{3}+1}{2\sqrt{2}}|+\rangle + \frac{\sqrt{3}-1}{2\sqrt{2}}|-\rangle.\end{aligned}$$

Now the amplitudes are easy to identify, and we can find the probabilities by taking their norm-squares. The probability of measuring $|+\rangle$ is

$$\left|\frac{\sqrt{3}+1}{2\sqrt{2}}\right|^2 = \frac{\sqrt{3}+2}{4} \approx 0.93,$$

Measurements in Other Basis



- And the probability of measuring $|-\rangle$

$$\left|\frac{\sqrt{3}-1}{2\sqrt{2}}\right|^2 = \frac{-\sqrt{3}+2}{4} \approx 0.07$$

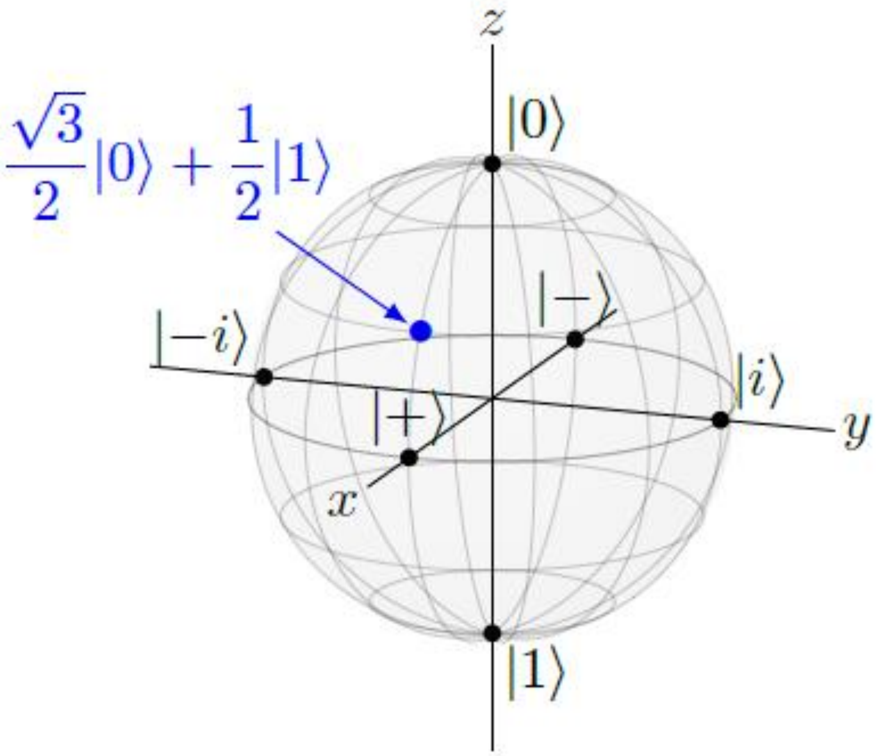
- This is consistent with the Bloch sphere, since the state is much closer to $|+\rangle$ than it is to $|-\rangle$

- Now what if we measure in the $\{|i\rangle, |-i\rangle\}$ basis?
- As we did before

$$|0\rangle = \frac{1}{\sqrt{2}}(|i\rangle + |-i\rangle), \quad |1\rangle = \frac{-i}{\sqrt{2}}(|i\rangle - |-i\rangle).$$

Substituting,

$$\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle = \frac{\sqrt{3}}{2}\frac{1}{\sqrt{2}}(|i\rangle + |-i\rangle) + \frac{-i}{\sqrt{2}}(|i\rangle - |-i\rangle)$$
$$= \frac{\sqrt{3}-i}{2\sqrt{2}}|i\rangle + \frac{\sqrt{3}+i}{2\sqrt{2}}|-i\rangle.$$



So, the probability of getting $|i\rangle$ is

$$\left|\frac{\sqrt{3}-i}{2\sqrt{2}}\right|^2 = \frac{3+1}{8} = \frac{1}{2},$$

and the probability of getting $|-i\rangle$ is

$$\left|\frac{\sqrt{3}+i}{2\sqrt{2}}\right|^2 = \frac{3+1}{8} = \frac{1}{2},$$

as expected.

Measurements in Other Basis

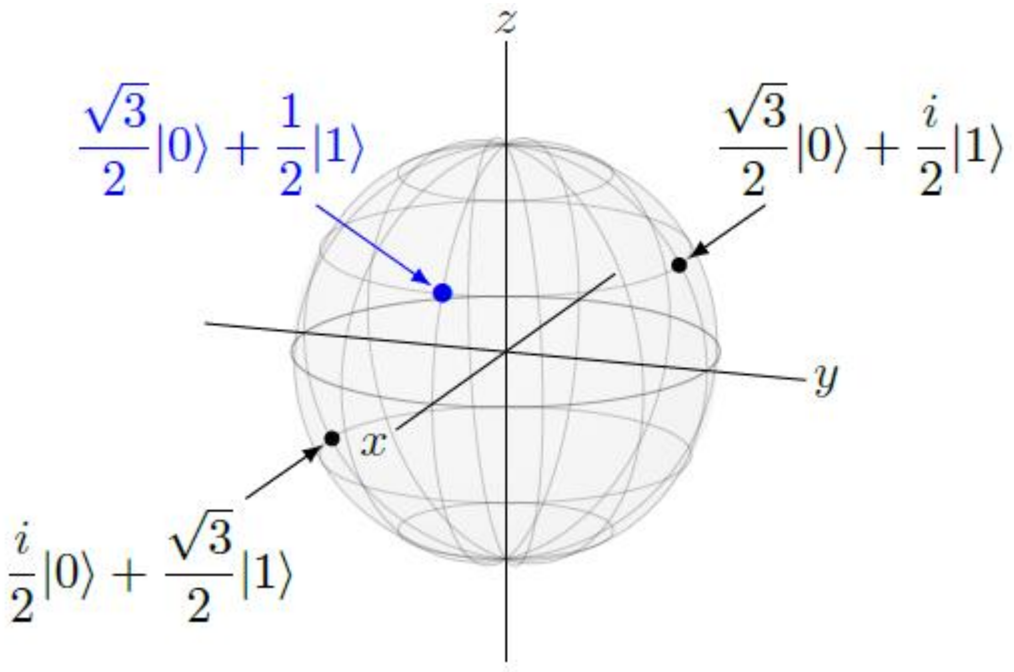
- Consider the following two states, which we will call $|a\rangle$ and $|b\rangle$

$$|a\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle \qquad |b\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

Since they are located on opposite points of the Bloch sphere, they are a basis.

Let us measure in this $\{|a\rangle, |b\rangle\}$ basis.

To calculate the precise numbers, we used the approach already seen before.



$$|0\rangle = \frac{\sqrt{3}}{2}|a\rangle - \frac{i}{2}|b\rangle, \quad |1\rangle = \frac{-i}{2}|a\rangle + \frac{\sqrt{3}}{2}|b\rangle.$$

Substituting, the state of our qubit is

$$\begin{aligned} \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle &= \frac{\sqrt{3}}{2}\left(\frac{\sqrt{3}}{2}|a\rangle - \frac{i}{2}|b\rangle\right) + \frac{1}{2}\left(\frac{-i}{2}|a\rangle + \frac{\sqrt{3}}{2}|b\rangle\right) \\ &= \frac{3-i}{4}|a\rangle + \frac{\sqrt{3}(1-i)}{4}|b\rangle. \end{aligned}$$

Taking the norm-square of each amplitude, the probability of getting $|a\rangle$ is

$$\left|\frac{3-i}{4}\right|^2 = \frac{9+1}{16} = \frac{5}{8},$$

and the probability of getting $|b\rangle$ is

$$\left|\frac{\sqrt{3}(1-i)}{4}\right|^2 = \frac{3}{8}.$$

Later, when we describe qubits in the mathematics of linear algebra, we will see another way to convert between bases.

Note on Single-Qubit Measurement

- *All measurement devices have associated bases*, and the measurement outcome is always one of the two basis vectors

- For this reason, whenever anyone says “measure a qubit," they must specify with respect to which basis the measurement takes place

- However, when we say “measure a qubit" without any further specification, we implicitly mean that the measurement is with respect to the standard basis or CBS $\{|0\rangle, |1\rangle\}$